Karlsruhe Institute of Technology

National High-Performance Computing Center

Machine learning in weather and climate research

StuMeTa Workshop 2026

May 14, 2026

Miriam Simm

Todays topics

- What is machine learning (ML)?
- Applications in weather and climate science
- ML algorithms
- Hands on session
- Concepts of machine learning (bonus)

What comes to mind when you think of ”ML in weather and climate research”?

Link to Mentimeter

Why is ML helpful for weather and climate research?

- Learn directly from observational data or reanalysis data
  $\rightarrow$ better predictions?
- ML models, once trained, are typically computationally cheaper to run than running traditional weather forecasting or climate models on supercomputers
  $\rightarrow$ faster or more resource-efficient predictions?
- Find new patterns and relationships in high-dimensional data
  $\rightarrow$ better predictions?

What is machine learning?

Artificial intelligence (AI): Sub-field of computer science, umbrella term for ”intelligent” algorithms that gather information from input data and process them (not precisely defined)

Machine learning (ML): Sub-field of AI, describes algorithms that learn to do a task based on ”experience” (data), without explicitly being programmed to do so

- Arthur Samuel (IBM/Stanford), 1959: ”*The field of study that gives computers the ability to learn without explicitly programmed*”

Neuronal networks (NNs): Algorithms that are inspired by neural connections in the human brain

- Consist of notes that are connected by weighted connections
- *Deep Learning:* NNs with many layers of internal neurons

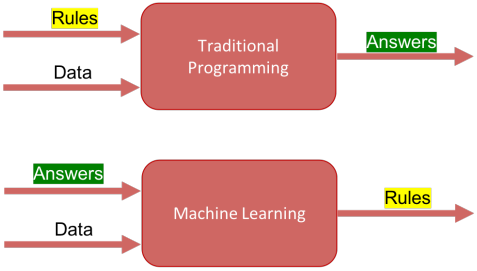
What is machine learning?

What?

- Extract information from data
- For data with input-output pairs (x, y), find the mapping from $x \to y$

How?

- Algorithm learns based on a pre-defined criterion and information content and structure of the data
- Algorithm does not model the solution but learns independently from repetion and feedback



(datalya.com [1])

What is machine learning?

Supervised Learning

- Regression
- Classification

Linear regression
Neuronal networks
Random forests
Decision trees
k-nearest neighbours

Unsupervised Learning

- Clustering
- Dimensionality-reduction

K-means clustering
PCA
Autoencoder

Reinforcement Learning

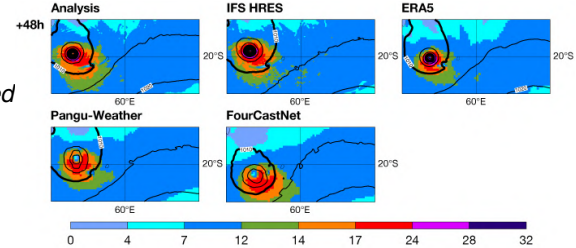
Monte Carlo methods

ML applications in weather and climate research

Link to Mentimeter

ML applications in weather and climate research

- Numerical weather prediction *(reinforcement learning)*
- Classification, e.g. of cloud types *(unsupervised learning)*
- Object identification, e.g. cyclones
- Anomaly detection, e.g. extreme events
- Parameterizations: speed-up, general improvement *(supervised learning)*



(ECMWF [2])

ML applications in weather and climate research

Huge amounds of data: observations, reanalysis data, model output

- Time series data $\rightarrow$ analogy to large language models (LLMs)
- Complexity and local effects can be captured effectively
- Requires less computational effort compared to traditional forecasting models

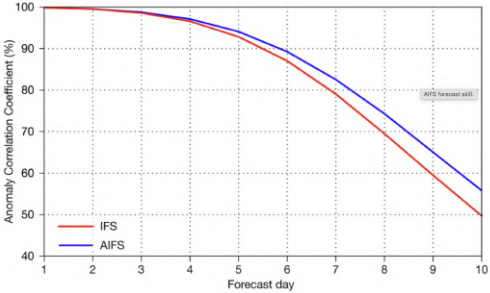
Is ML/AI the future of weather forecasting?

ECMWF's ensemble AI forecasts become operational

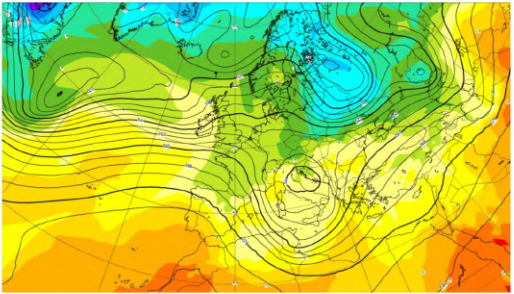
1 July 2025

Share

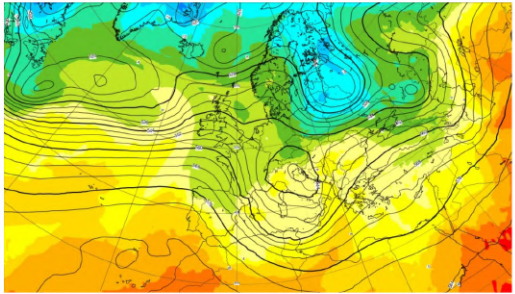
ECMWF has taken the ensemble version of the Artificial Intelligence Forecasting System (AIFS) into operations today, 1 July 2025, to run side by side with the traditional physics-based Integrated Forecasting System (IFS) to advance numerical weather prediction.



Is this a machine learning forecast? (24 hours)



ECMWF AIFS Single - ML

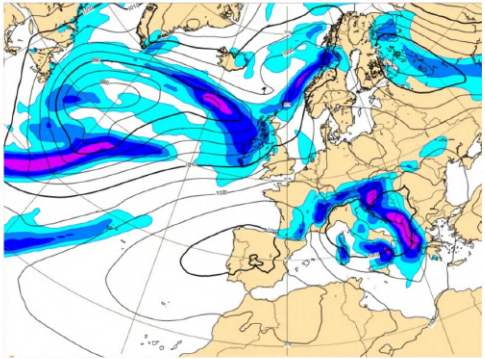


ECMWF Control Forecast / IFS – physics-based

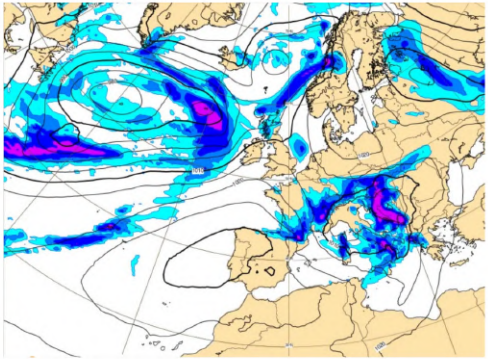
500 hPa geopotential height (contours) and 850 hPa temperatures (colours)
Starting point: 19. February 2026, 00 UTC

Credits: Peer Nowack, KIT/ECMWF

Is this a machine learning forecast? (24 hours)



ECMWF AIFS Single – ML



ECMWF Control Forecast / IFS – physics-based

Mean sea level pressure (contours) and precipitation (colours)
Starting point: 19. February 2026, 00 UTC

Credits: Peer Nowack, KIT/ECMWF

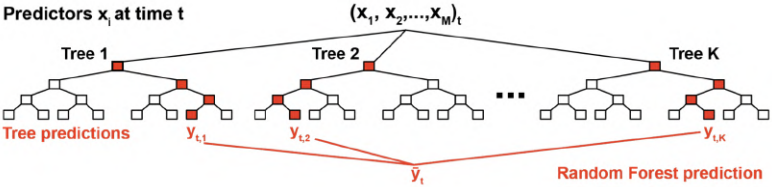
ML algorithms

Decision trees

- Simple ML algorithm for classification and regression
- Consists of *nodes* and *leaves*
- Tree ”grows” with the CART algorithm
- Finds optimal features (predictors) and thresholds that maximize the information gain at each node ($\rightarrow$ Gini impurity)
- Prediction: mean of all samples in a leaf
  - $\rightarrow$ classification: probability or class
  - $\rightarrow$ regression: value
- Easy to understand and implement

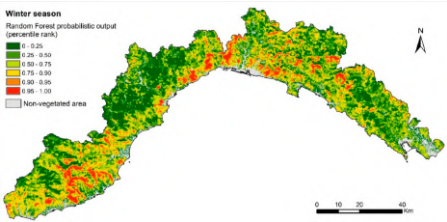
Example: cloud type classification *Huertas-Tato et al. [3]*

Random forests



(T.T-Prof. Dr. Peer Nowack, KIT)

- Example for **ensemble methods**: improved prediction by combining multiple individual models (decision trees)
- Further improved by introducing randomness to the algorithm (e.g. *bagging*)



Example: probability of wild fires *(Tonini et al. [4])*

Linear regression

Linear function of independent variables $\vec{x} = (x_1, \dots, x_n) \in \mathbb{R}^n$ to make predictions y of the target $t \in \mathbb{R}$

$$y = \sum_j w_j x_j + b$$

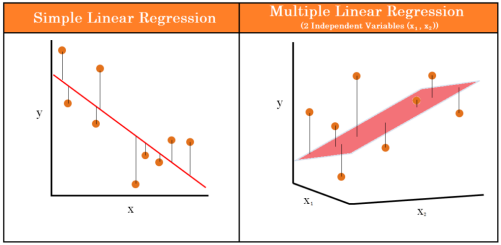
w : Slope $\xrightarrow{\text{ML}}$ *weights*

b : y-intercept $\xrightarrow{\text{ML}}$ bias

x : Independent variable $\xrightarrow{\text{ML}}$ input-*feature*, predictor

t : Target $\xrightarrow{\text{ML}}$ *target*, label

y : Output value, prediction



(Antony Christopher (medium.com) [5])

One feature, one output value:

$$y = wx + b$$

Multiple features:

$$y = \vec{w}^\top \vec{x} + b \quad \vec{w}, \vec{x} \in \mathbb{R}^n$$

Generalization for multiple output values:

$$\vec{y} = W^\top \vec{x} + \vec{b} \quad \vec{y}, \vec{b}, \vec{x} \in \mathbb{R}^n, W \in \mathbb{R}^{m \times n}$$

Linear regression

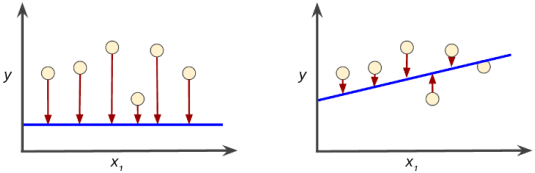
Parameter estimation $\xrightarrow{\text{ML}}$ **training**: find good values for weights and biases from labelled data

Method of least-squares: mean squared error for N data samples

$\xrightarrow{\text{ML}}$ Example for a **loss function**: measures how good a prediction is

$$\mathrm{MSE} = \frac{1}{N}\sum_{j=1}^{N}\left(y_j - t_j\right)^2$$

$$= \frac{1}{N}\sum^{N}(\text{prediction} - \text{target})^2$$



(Aspersh Upadhyay (medium.com) [6])

Find a model that minimizes the loss $\mathcal{L}(\vec{w}, b)$

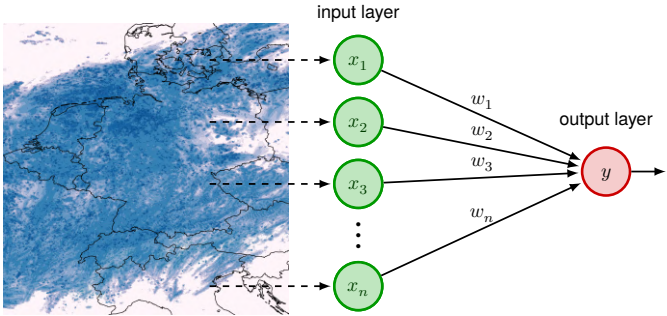
↪ Exact solution: derivative $= 0$

↪ Generally: an exact solution might not exist or is hard to find (non-linearity) $\rightarrow$ **gradient descent**

What are neural networks?

Definition: Algorithms that are inspired by the structure of the human brain and can model complex relationships

- Consist of an *input layer*, a number of *hidden layers* and an *output layer*, each of which consist of *(nodes)*
- Nodes are connected with weights and biases



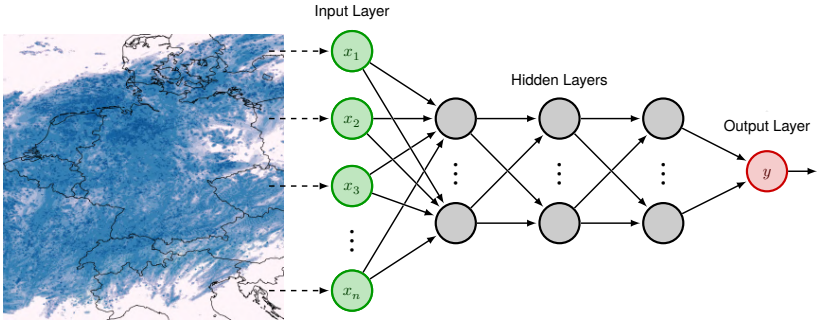
Recap: linear regression

$$y = w_1 x_1 + x_2 x_2 + \cdots + w_n x_n + b$$

$$\vec{y} = W^\top \vec{x} + \vec{b}$$

What are neural networks?

NNs need to model complex, non-linear relationships: add *hidden layers*



$$y = W^{(3)}(W^{(2)}(W^{(1)}\vec{x}))$$
$$W^{(1)} \in \mathbb{R}^{n\times k},$$
$$W^{(2)} \in \mathbb{R}^{k\times l},$$
$$W^{(3)} \in \mathbb{R}^{l\times m}$$

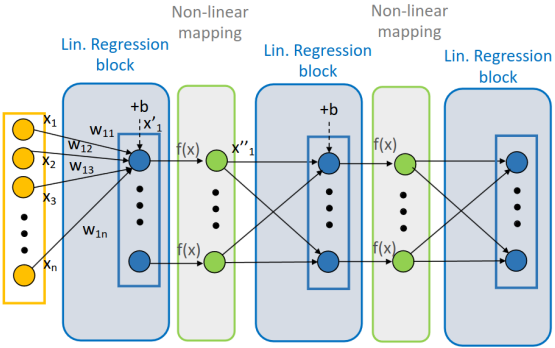
- Output of each node: regression with the node from the previous layer
- Each node has its own regression parameters
- However: result is still linear

What are neural networks?

Add non-linearity between each two layers:
activation function f

$$y = W^{(3)} f(W^{(2)} f(W^{(1)} \vec{x}))$$

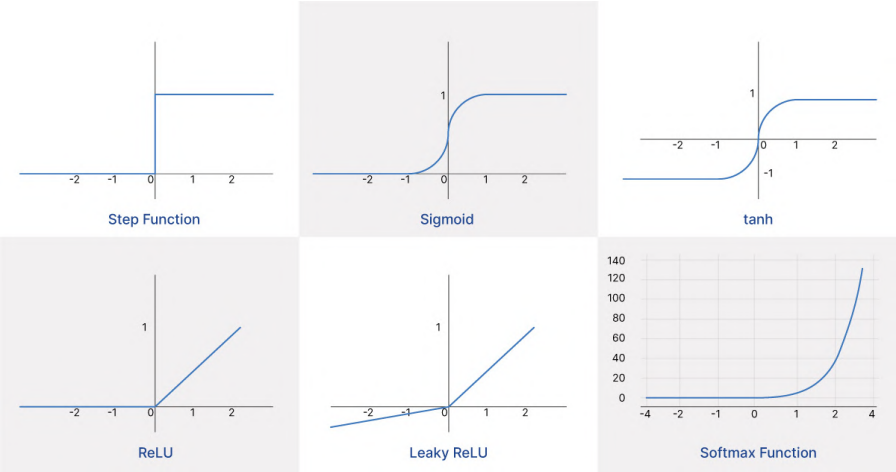
- Determine the output of a node based on the input
- The NN is now able to model complex relationships
- The choice of activation function is important



(Ryan Lagerquist, Imme Ebert-Uphoff (CIRA) [7])

Neural networks consist of a number of linear transformations *(→ layers)* with non-linear activation function in between layers

Activation functions

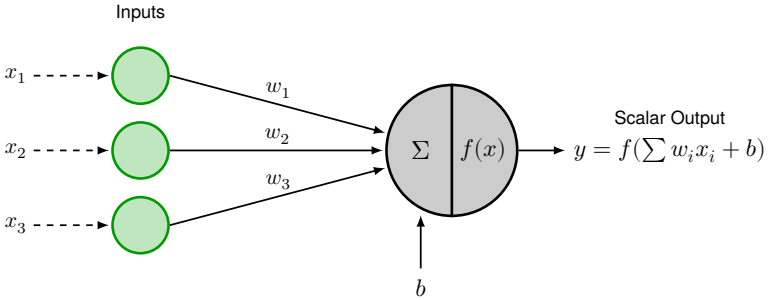


(shiksha.com [8])

The artificial neuron

Building block of a neural network, consists of:

- Input values *(features)* x
- Weights w for each input, which determine the degree of influence of each input
- Bias b, shifts the value of the weighted sum
- Activation function f, processes the weighted sum of the inputs and the bias and generates the output



How do NNs learn?

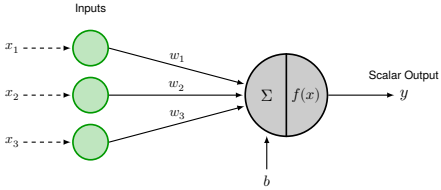
Training: Adjust the weights and biases so that the error in the output is minimised
$\rightarrow$ gradient descent

Parameters: are learned during model training

- Coefficients in linear regression
- Weights in a neural network

Hyperparameters: are selected before training and are not part of the learning process

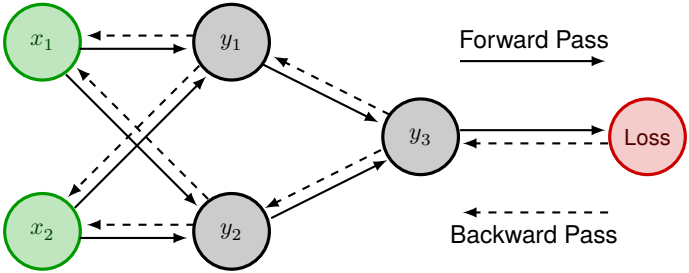
- Number of nodes and layers in a NN
- *Learning rate* α
- Activation function f
- ...



Backpropagation

Training algorithm for neural networks with multiple hidden layers:

- *Forward Pass*: Input data passes through the neural network, prediction and loss are calculated
- *Backward Pass*: Gradient descent, recursive application of the chain rule
- Parameter update



Gradient descent

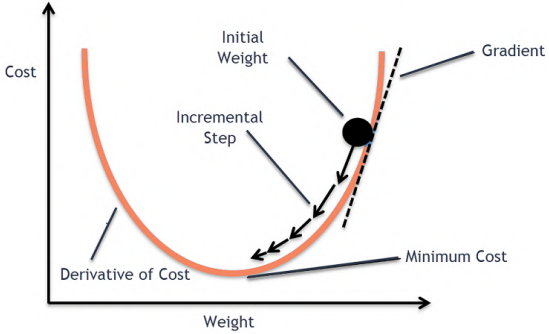
Adjusting the model's parameters (weights and biases) in the direction of the steepest descent of the loss function

→ **Iterative algorithm**: repeated parameter updates until the model is sufficiently accurate

1. Initialise the parameters with sensible default values (e.g. all set to zero or random numbers)
2. *Forward pass*
3. Calculate the gradient of the loss function at each node (recursively)
4. Update the parameters accordingly, multiplied by the **learning rate** α (typically $\alpha \in (0, 1]$)

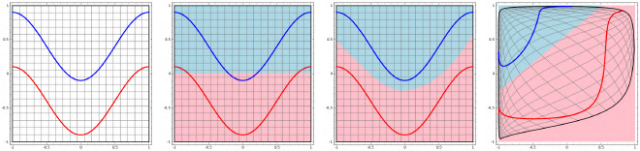
$$w_{ij} \leftarrow w_{ij} - \alpha \frac{\partial \mathcal{L}}{\partial w_{ij}}$$

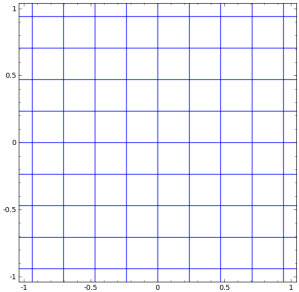
5. Repeat



(analyticsvidhya.com [9])

Example for classification





(Christopher Olah [10])

Example[1]

$$\tanh(Wx + b)$$

1. Linear transformation using the weight matrix W
2. Translation with bias vector b
3. pointwise application of the activation function $\tanh$

[1] For more mathematical background **Here** is a very interesting blog post about neural networks and topology

Input data

Three datasets are required[2]:

Training data to learn the model parameters
Validatiomn data for determining and adjusting the hyperparameters
Test data to evaluate model performance on data unseen during training

(Ryan Lagerquist, Imme Ebert-Uphoff (CIRA) [7])

- Datasets must be mutually exclusive and independent
- Be cautious of **temporal autocorrelation**: if time series data are shuffled at random, they are not independent but highly correlated

[2] Unless cross-validation is used, then only training and test datasets are needed

Evaluation: performance metrics

Applying the model to the validation data gives an indication of how well the model will perform in real-world applications and on the test set

- Classification: accuracy, precision, recall, F_1-score, …
- Regression: mean absolute error (MAE), mean squared error (MSE), root mean square error (RMSE), R^2-score, …

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|$$

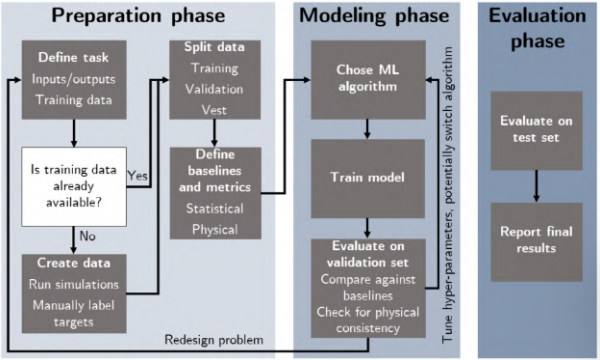
$$R^2 = 1 - \frac{\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{N}(y_i - \bar{y})^2}$$

Hands on session

The ML workflow

Before starting with ML, ask yourself

- Why do I want to use ML?
- What kind of data do I have? How large is my dataset, and do I need to process the data in any way beforehand?
- Which algorithm is fitting my task best?
- Can I break the problem down into smaller sub-problems that can then be solved using machine learning?



(Beucler et al. [11])

Hands-on session

Link to Google Colab

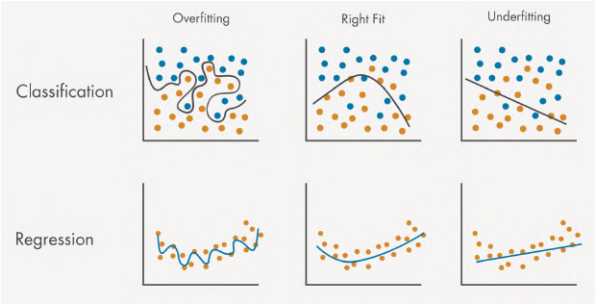
Link to Github Repository

Concepts of machine learning

The "Bias-variance trade-off"

Bias: Describes the error that occurs when a complex problem is described by a simpler model

Variance: describes the error induces by too high sensitivity to small fluctuations



(Everton Gomede (medium.com) [12])

High bias: Underfitting

→ Complexity is not captured sufficiently

→ The algorithm fails to identify relevant relationships between features and targets and is unable to capture the variance in the data

High variance: Overfitting

→ Sensitivity to small fluctuations

The ”Bias-variance trade-off”: mitigation strategies

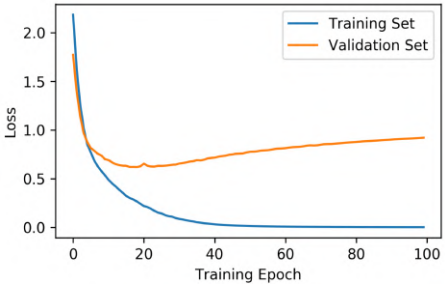
High bias *(Underfitting)*: Build a more complex model (mode features, more training epochs, larger trees, …)

High variance *(Overfitting)*: reduce model complexity

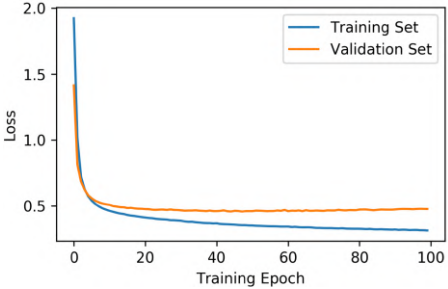
- **Feature selection**:Select only the most relevant features and reduce the complexity of the model
- Use a larger training set
- **Regularization** (L_1- and L_2-regularization), dropout, cross-validation

Learning curves

Indicate over- and underfitting: Performance on the training/validation dataset plotted against a suitable hyperparameter (in this case, the number of training epochs)



The learning curve shows overfitting *(Prof. Dr. Rhode, TU Dortmund [13])*

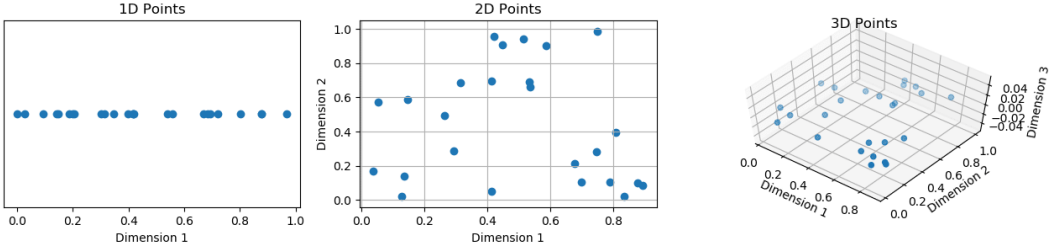


The learning curve demonstrates the desired behaviour *(Prof. Dr. Rhode, TU Dortmund [13])*

The "Curse of Dimensionality"

The distances between data points grow exponentially with the number of dimensions *(features)*

- Training data points are widely scattered → pattern recognition is difficult
- Greater complexity, combined with increased computational effort
- Can induce overfitting



(aiaspirant.com [14])

Summary

- Machine learning refers to algorithms that learn from data and are able to make predictions or decisions based on data
- Machine learning is a powerful and effective method that is already widely used in weather and climate science $\rightarrow$ a rapidly growing field
- However, machine learning is not the best approach for every problem and is not a 'one-fits-all' solution, a thorough analysis of the problem is crucial

References

[3] J. Huertas-Tato et al. “Automatic Cloud-Type Classification Based On the Combined Use of a Sky Camera and a Ceilometer.” In: Journal of Geophysical Research: Atmospheres 122.20 (2017), pp. 11, 045–11, 061. DOI: https://doi.org/10.1002/2017JD027131. eprint: https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1002/2017JD027131. URL: https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1002/2017JD027131.

[4] Marj Tonini et al. “A Machine Learning-Based Approach for Wildfire Susceptibility Mapping. The Case Study of the Liguria Region in Italy.” In: Geosciences 10.3 (2020). ISSN: 2076-3263. DOI: 10.3390/geosciences10030105. URL: https://www.mdpi.com/2076-3263/10/3/105.

[7] Imme Ebert-Uphoff Ryan Lagerquist. CIRA Short Course on Machine Learning for Weather and Climate. URL: https://docs.google.com/document/d/1SPNxZrbHMaIEaS2dbntDow9x_tgSuFTUTOugfa2NuRo/edit (visited on 07/13/2024).

[10] Christopher Olah. Neural Networks, Manifolds, and Topology. URL: http://colah.github.io/posts/2014-03-NN-Manifolds-Topology/ (visited on 07/13/2024).

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[18] Prof. Dr. Wolfgang Rhode. Vorlesung Statistische Methoden in der Datenanalyse.

[19] Leo Breimann et al. CART: Classification and Regression Trees. Wadsworth:Belmont, CA, 1984.

Image Sources

[1] Machine learning vs. traditional programming paradigm. URL: https://datalya.com/blog/machine-learning/machine-learning-vs-traditional-programming-paradigm (visited on 07/13/2024).

[2] The rise of machine learning in weather forecasting. URL: https://www.ecmwf.int/en/about/media-centre/science-blog/2023/rise-machine-learning-weather-forecasting (visited on 07/13/2024).

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